Macroeconomics

Time: 120 minutes

Total: 12 points

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Main Exam

This exam contains two parts.

Using AI is strictly prohibited. Reviewing past lecture notes are possible.

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1. (5 points) Consider the following Ramsey-style dynamic optimization problem

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

where

$$u(c_t) = \ln(c_t - \theta), \qquad c_t > \theta$$

subject to

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t.$$

given $\beta, \delta \in (0, 1)$ and $\theta > 0$.

   (a) Write the Lagrangian and derive the Euler equation of the optimal consumption-saving.

   (b) Compute the intertemporal elasticity of substitution $IES = -\dfrac{u'(c)}{c\,u''(c)}$. Explain the meaning and its implications for a poor household with low c versus a rich household with high c.

   (c) Derive the change in consumption in terms of θ and the model parameters. What does a higher θ imply for the consumption smoothing behavior?

   (d) Derive the steady state level of consumption c^*. Assume that $f(k) = Ak^\alpha$. What is the steady state level of capital k^*? Does it depend on parameter θ?

   (e) Assume that the government makes a one-time positive transfer to households at $t = 0$. What would be the likely response of poor household (low c) and rich household (high c) regarding consumption now (and saving for later)?

2. (7 points) We want to study the availability and effectiveness of contraceptives (các biện pháp phòng tránh thai không mong muốn) on fertility and the macroeconomy.

Production

A representative firm produces output Y_t using capital K_t and labor L_t with a Cobb–Douglas technology:

$$Y_t = AK_t^{\alpha} L_t^{1-\alpha}, \qquad \alpha \in (0,1), \;\; A > 0. \tag{1}$$

Capital depreciates fully within one period ($\delta = 1$). Labor is supplied by young workers. Each young worker supplies one efficiency unit, net of child-rearing time.

Demography

Time is discrete, $t = 0, 1, 2, \ldots$. In each period N_t identical young agents are born; each lives for three periods—childhoold (period $t+1$), young (period t) and old (period $t+1$). No economic decision is made during childhood. People work only during youth.

The population of young working agents evolves according to

$$N_{t+1} = n_t N_t, \tag{2}$$

where n_t is the *realized* fertility rate of a young household in period t.

Fertility

Realized births consist of two components:

$$n_t = n_t^d + (1-\kappa)u, \tag{3}$$

where

- $n_t^d \geq 0$ is *desired* (or intended) fertility, chosen by the household;
- $u > 0$ is the *unintended fertility shifter*, a biological parameter capturing the maximum number of unintended births in the absence of contraception;
- $\kappa \in [0,1]$ is *contraceptive effectiveness*: the fraction of unintended births that contraception prevents.

When $\kappa = 0$, no contraception is used and realized fertility is $n_t = n_t^d + u$; when $\kappa = 1$, contraception is perfect, i.e., the household can perfectly control births and $n_t = n_t^d$. When $\kappa < 1$, the contraceptive is not perfect, resulting in realized births larger than intended.

Preferences

An important assumption is that households derive utility only from *desired* children, while the budget constraint is determined by *realized* births.

Each young household maximizes lifetime utility

$$U_t = \ln c_t^y + \beta \ln c_{t+1}^o + \gamma \ln n_t^d, \tag{4}$$

where $\beta, \gamma \in (0,1)$ and:

- c_t^y is young-age consumption,
- c_{t+1}^o is old-age consumption,
- $\beta > 0$ is the subjective discount factor,

- $\gamma > 0$ is the weight on desired children in utility.

A young household supplies one unit of labor inelastically and receives the competitive wage w_t. Raising children requires a time cost: each child (desired or unintended) consumes a fraction $\phi \in (0,1)$ of the household's time endowment. Thus total labor income net of child-rearing is

$$(1 - \phi n_t)w_t = \big(1 - \phi n_t^d - \phi(1-\kappa)u\big)w_t. \tag{5}$$

The young allocate this income between consumption and saving:

$$c_t^y + s_t = (1 - \phi n_t)w_t, \tag{6}$$
$$c_{t+1}^o = R_{t+1}s_t, \tag{7}$$

where $s_t \geq 0$ denotes saving and R_{t+1} is the gross return on capital in $t+1$ (the interest factor).

Answer the following questions.

(a) Write the firm's profit maximization problem and derive the optimal capital rental rate R_t and wage rate w_t.

(b) Solve the household problem that maximizes lifetime utility by choosing the optimal c_t^y, c_{t+1}^o, n_t^d. In other words, household maximizes (4) with respect to (5),(6),(7). You may want to combine the three contraints into one single budget constraint.

(c) Given the optimal desired fertility n_t^d solved above, derive the saving function s_t and the realized fertility n_t.

(d) Is it possible for the desired fertility to fall to zero? Find the condition for $n_t^d > 0$ always.

(e) The capital market clears according to

$$K_{t+1} = N_t s_t$$

Write the law of motion for capital per worker k_{t+1}.

(f) What are the steady-state values of capital per worker, desired and realized fertility?

(g) What happens to the steady-state values of capital per worker, desired fertility, and realized fertility when there is an increase in κ? Show it. Can you give an intuitive explanation of such changes?

END OF EXAM
